Supplementary File 7: Data Collection and Harmonization

Due to the different provincial division systems in each country, this study compares case studies at different administrative levels: the NUTS3 Level (counties) in Germany (DE935 Lüneburg and DE934 Lüchow-Dannenberg) and the LAU1 Level (districts) in Türkiye (Lüleburgaz and Babaeski).

Data collection for this cross-country comparison encompassed roughly 50 variables for each case (see Repository). Data was gathered from multiple sources:

- Demographic data was obtained from Eurostat (Germany) and TÜİK (Türkiye).
- Land-cover data was gathered from Corine (Corine, 2018).
- Spatial data (administrative boundaries, roads, settlements, and POI/facilities) was collected from Open Street Map (OSM, 2019).
- Missing data for POIs was supplemented via Google Maps.
- Calculated data included variables such as distance to the district center for each settlement.

Given the different sources, administrative levels (NUTS-3 vs. LAU-1), statistical years, and definitions, a multi-step harmonization process was required. This was essential for creating *functionally equivalent* variables for the cross-country comparison.

Our method involved:

1. Temporal & Definitional Alignment: (e.g., selecting closest years/proxies).
2. Categorical Alignment: (e.g., unified re-classification).
3. Scale Alignment: (i.e., standardizing all variables to Z-scores, as detailed in the Statistical Analysis section).

Table S 7 provides a complete, variable-by-variable explanation of these harmonization steps and their justifications.

Table S 7: Cross-Country Data Harmonization Framework

Variable Category	Variable Name	Germany (Lüneburg)	Türkiye (Trakya)	Harmonization Step Taken
Socio-Demographic	Population	**Source:** Destatis / Eurostat **Year:** 2018 **Definition:** Total registered inhabitants.	**Source:** TÜİK **Year:** 2018 **Definition:** Total registered inhabitants.	1. Data was temporally aligned. 2. Standardized to Z-score.
Socio-Demographic	Elder Rate	**Source:** Destatis / Eurostat **Year:** 2011 **Definition:** % of pop. > 65.	**Source:** TÜİK **Year:** 2000 **Definition:** % of pop. > 65.	1. Acknowledged 11-year temporal gap is unavoidable. 2. Used as best available proxy for elder population. 3. **Note:** Data for Lüleburgaz district the entire *elder* variable column was missing. 4. Standardized to Z-score.

Socio-Demographic	Pop. Change Rate	**Source:** Destatis / Eurostat **Year:** 2016-2018 **Definition:** % change.	**Source:** TÜİK **Year:** 2017-2018 **Definition:** % change.	1. Acknowledged minor difference in time period. 2. Standardized to Z-score.
Socio-Demographic	Economic Health	**Source:** Destatis / Eurostat **Variable:** Unemployment rate **Year:** 2018	**Source:** Ministry of Development **Variable:** Dev_ind_2004 **Year:** 2004	1. Acknowledged major definitional and temporal differences. 2. Used as best available proxies for economic status. 3. Standardized to Z-score.
Land Cover	Land Cover	**Source:** CORINE **Year:** 2018	**Source:** CORINE **Year:** 2018	1. Source/Year identical. 2. Re-classified raw CLC codes into 6 aggregated categories: • Urban (111-112) • Industrial (121-124) • Mine (131-133) • Agri (2xx) • Forest/Green Urban (3xx) • Water/Wetlands (4xx/5xx).
Facility Counts	Facilities (POI)	**Source:** OpenStreetMap **Year:** 2022	Source: OpenStreetMap Year: 2022	1. Source/Year identical. 2. Aggregated raw fclass tags into 6 functional categories: • Public (Schools, Health, Gov) • Commercial (Retail, Banks) • Recreation (Parks, Culture) • Transport (Stops, Stations) • Environmental (Water, Waste) • Religious (Full code list is given below).
Spatial Network	Road Network	**Source:** OpenStreetMap **Year:** 2021	**Source:** OpenStreetMap **Year:** 2021	1. Data source and year were identical. 2. Road types are categorized under 5 groups, Railways (5), International motorway and trunk roads (4), Primary national roads (3), Secondary Regional Roads (2) Tertiary local roads (1)
Spatial Network	Admin. Boundaries	**Source:** OpenStreetMap **Year:** 2022	**Source:** OpenStreetMap **Year:** 2022	1. Data source and year were identical.

CORINE Classification

1. 11_URBAN: "Code_18" IN('111', '112')

2. 12_INDUSTRIAL: "Code_18" IN('121', '122', '123', '124')

3. 13_MINE: "Code_18" IN('131' , '132' , '133')

4. 2_AGRI: "Code_18" IN('211' , '212' , '213' , '221' , '222' , '223' , '231' , '242' , '243')

5. 3_FOREST and 14_GREEN_URBAN: "Code_18" IN('311' , '312' , '313' , '321' , '323' , '324' , '331' , '333' , '334' , '142')

6. 4_WETLANDS and 5_WATERBODIES: "Code_18" IN(‘411’, ‘412’, ‘421’, ‘422’, '423', ‘511’, ‘512’, ‘521’, ‘522’, '523')

OSM Classification:

Public Facilities

"type" IN ("fclass" IN ('childcare' , 'clinic' ,'college' , 'dentist', 'embassy' ,'hospital' , 'library' , 'language_school', 'preschool' , 'school' ,'social_facility' , 'townhall' ,'university', 'retirement_home', 'services' , 'doctors', 'kindergarten', 'nursing_home' , 'police', 'post_office' , 'public_bookcase' , 'driving_school' , 'fire_station', 'animal_shelter' , 'music_school', 'community_centre', 'public_building', 'town_hall')

Recreation- Attraction Facilities

"type" IN "fclass" IN ('archaeological_site' , 'arts_centre' , 'artwork' , 'attraction' ,

'festival_grounds' , 'gallery' , 'heritage' , 'monument' , 'museum' , 'theme_park' ,'viewpoint' , 'zoo' , 'camp_site' , 'caravan_site' , 'cinema' , 'information' , 'memorial' ,'picnic_site' , 'rest_area' , 'theatre', 'clock', 'community_centre', 'communications_t', 'communication_tr', 'public_bath', 'public_bookcase' , 'rest_area', 'sauna', 'statue', 'swimming_pool', 'fountain' , 'ruins', 'sauna', 'statue', 'telephone', 'tomb', 'tower' , 'viewpoint', 'cabin', 'biergarten', 'garden_centre', 'golf_course', 'stadium', 'sports_centre', 'tourist_info')

Commercial Facilities:

"fclass" IN ('atm' , 'bakery' ,'bank','bar' , 'beauty_shop', 'beverages' , 'bicycle_shop' , 'bookshop' , 'butcher' , 'cafe' , 'car_rental','car_wash' , 'casino', 'computer_shop', 'fast_food', 'food_court','guest_house' , 'gift_shop', 'hairdresser' ,'hostel', 'hotel' , 'jeweller', 'kiosk', 'laundry', 'mobile_phone_shop' , 'motel' , 'optician' , 'outdoor_shop' , 'pharmacy' ,'ice cream' , 'ice_cream' ,'internet_cafe', 'marketplace', 'motel', 'nightclub', 'pub', 'restaurant', 'studio', 'swingerclub', 'shoe_shop' , 'sports_shop' , 'supermarket' , 'toy_shop' , 'video_shop' ,'taxi','works', 'advertising', 'baking_oven' , 'canteen', 'fuel')

Environmental Facilities:

"fclass" IN ('monitoring_stati' ,'fountain', 'drinking_water' ,'water_well', 'watermill', 'water_point', 'wastewater_plant', 'water_tap' , 'water_tower' , 'water_works' , 'watering_place' ,'waste_disposal', 'waste_basket', 'recycling', 'reservoir_covere', 'windmill', 'recycling_paper', 'recycling_metal', 'recycling_glass', 'recycling_clothes')

Transportation Facilities:

"fclass" IN (‘railway_station’ ,’ railway_halt’ ,’ tram_stop’ , ‘bus_stop’, ‘bus_station’, ‘taxi_rank’, ‘airport’, ‘airfield’, ‘helipad’, ‘apron’, ‘ferry_terminal’)

Religious Facilities:

"fclass" IN ('place_of_worship')